33 · The ovarian and uterine cycles

The words of this track, as it says them. Bold brown is the examiner.

The ovarian and uterine cycles. Six marks.

Describe the phases of the ovarian and uterine cycles in the female, and how the hormones link them.

She posted no model answer for this one, so this answer is built straight from her Repro Female deck. It's six marks, and it comes back in the final, in the fertility case, so it's worth the effort.

Two cycles run side by side: the ovary's, and the uterus's. Day one is the first day of the period.

The hook. The ovary: grow, pop, yellow body, white scar. The uterus: shed, rebuild, get ready.

The ovary first. The follicular phase, days one to fourteen. FSH from the anterior pituitary makes a follicle grow: primordial, primary, secondary, and then a mature Graafian follicle. Its granulosa cells secrete oestrogen.

Ovulation, mid-cycle, at about day fourteen. A surge of LH makes the Graafian follicle rupture and release the secondary oocyte. That's the pop.

The luteal phase, days fourteen to twenty-eight. The ruptured follicle becomes the corpus luteum, the yellow body, which secretes progesterone, and some oestrogen. If there's no pregnancy, it breaks down after about ten days into the corpus albicans, the white scar.

Now the uterus. The menstrual phase, days one to five. Progesterone falls, the spiral arteries spasm, and the stratum functionalis of the endometrium is shed.

The proliferative phase, days six to fourteen. Rising oestrogen from the growing follicle rebuilds the endometrium.

The secretory phase, days fifteen to twenty-eight. Progesterone from the corpus luteum turns the endometrium into a gland-rich lining, ready for implantation. If no embryo implants, the corpus luteum breaks down, progesterone falls, and menstruation starts again. That fall in progesterone is the link that restarts both cycles.

The traps. The follicular phase goes with FSH and oestrogen. The luteal phase goes with the corpus luteum and progesterone. Second: proliferative is oestrogen, and secretory is progesterone. Third: only the stratum functionalis is shed. The stratum basalis stays, and grows the next one.

Once more. Both cycles.

The ovarian cycle. In the follicular phase, FSH makes a follicle grow into a Graafian follicle, which secretes oestrogen. At ovulation, around day fourteen, an LH surge ruptures it and

releases the secondary oocyte. In the luteal phase the corpus luteum secretes progesterone, and without a pregnancy it becomes the corpus albicans. The uterine cycle. In the menstrual phase, progesterone falls and the stratum functionalis is shed. In the proliferative phase, oestrogen rebuilds it. In the secretory phase, progesterone readies it for implantation. With no implantation, progesterone falls, and menstruation starts again.